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Design and Development of a 2.5 kWh Lithium Battery for High-Capacity Energy Storage Solution in Modern Applications

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Abstract: Frequent power outages and an unstable electricity supply continue to affect residential buildings, workshops, and small businesses, especially in developing economies. Conventional backup systems based on lead-acid batteries are often limited by low energy density, poor cycle life, excessive maintenance, and unstable voltage response under heavy loads. Although lithium-based alternatives have emerged as promising substitutes, many low-cost systems still suffer from poor thermal management, weak battery management integration, inaccurate state-of-charge monitoring, and rapid performance degradation during repeated charge-discharge cycles. This study presents the design, fabrication, and experimental validation of a 24 V, 2.5 kWh lithium iron phosphate ($LiFePO_4$) battery energy storage system integrated with an intelligent monitoring and protection architecture for modern backup power applications. The proposed system employed an 8S1P configuration using 100 Ah $LiFePO_4$ cells, a smart battery management system (BMS), a 30 A intelligent charger, and an Arduino-based real-time display interface for voltage and current monitoring. Experimental results showed that the battery absorbed approximately 23.5 A at nearly 25% state of charge, indicating excellent charge acceptance and stable regulation. During discharge testing, the system delivered 2.56 kWh while maintaining a gradual voltage reduction from 29.2 V to 23.5 V, demonstrating improved energy retention and output stability. The developed system provides a reliable, safe, and scalable alternative to conventional backup storage technologies. Its practical value includes improved household energy reliability, reduced maintenance requirements, enhanced operational lifespan, and suitability for renewable energy integration in homes, offices, and microgrid applications.

Keywords: $LiFePO_4$ battery; Energy storage system; Battery management system; Intelligent monitoring; Lithium battery fabrication; Embedded energy monitoring

1 Introduction

The growing dependence on electrical energy for residential, educational, and commercial activities has increased the demand for reliable and efficient energy storage systems. In many regions with unstable power infrastructure, frequent outages have forced households and small businesses to depend heavily on backup power solutions. However, conventional lead-acid batteries commonly used for these applications suffer from several technical limitations, including short lifespan, slow charging rate, poor depth-of-discharge capability, electrolyte leakage, heavy weight, and unstable voltage delivery under varying loads. These problems reduce system reliability and increase maintenance and replacement costs [1–3].

Although lithium-ion batteries provide better energy density and operational efficiency, many commercially available low-cost lithium backup systems still face major challenges such as inadequate battery management coordination, inaccurate state-of-charge estimation, thermal instability, poor cell balancing, and weak protection against overcharge and deep discharge conditions. These limitations often result in premature battery degradation and unreliable system performance during prolonged operation. Therefore, there is a need for a smart, durable, and efficiently monitored lithium battery system capable of delivering stable output power, enhanced protection, and improved operational lifespan for modern energy storage applications. Lithium iron phosphate ($LiFePO_4$) batteries have recently gained considerable attention due to their superior thermal stability, high cycle life, safety characteristics, and environmental friendliness compared to conventional lithium-ion chemistries. Recent studies have shown that $LiFePO_4$ systems provide improved operational safety and enhanced lifecycle performance when combined with intelligent battery management and monitoring algorithms [4–6].

Despite significant advancements in lithium-based energy storage technologies, several technical challenges remain insufficiently addressed in existing backup power systems. Many commercially available lithium battery systems primarily focus on energy storage capacity and battery protection, with limited emphasis on integrated real-time monitoring, intelligent supervisory control, and user-centered operational visibility. Furthermore, a large proportion of reported studies concentrate on advanced battery management algorithms, state-of-charge estimation techniques, or electric vehicle applications, often requiring sophisticated hardware, high computational resources, and increased implementation costs that may not be practical for residential and small-scale commercial users [4, 6].

Recent studies have investigated intelligent battery management systems (BMS) and energy storage integration for microgrids and renewable energy applications. For example, Kandari et al. [7] reviewed strategies for integrating battery energy storage systems into microgrids, emphasizing system-level control and energy optimization. Similarly, Kechida et al. [8] proposed smart control approaches for renewable-energy-based hybrid systems, focusing on energy management and operational coordination. Pang et al. [9] focused on improving the accuracy of state-of-charge estimation for $LiFePO_4$ batteries using advanced computational models, while Kokoette [5] examined battery management challenges in aerospace applications. Although these studies contribute significantly to battery technology development, they provide limited practical implementation details for low-cost, compact, and standalone backup energy storage systems suitable for households, offices, and workshops.

In addition, most existing backup battery solutions either rely solely on standard BMS for protection or employ monitoring architectures that do not provide continuous real-time visualization of battery operating conditions. Consequently, users often have limited awareness of charging status, load behavior, battery health, and fault conditions, which can lead to inefficient operation, premature battery degradation, and reduced system reliability.

To address these limitations, this study presents a fully integrated 24 V, 2.5 kWh $LiFePO_4$ battery energy storage system that combines battery protection, intelligent charging, embedded monitoring, and real-time supervisory control within a single compact architecture. Unlike previous studies that primarily focus on algorithm development or large-scale energy management frameworks, the proposed system emphasizes practical implementation, affordability, and operational transparency. The major contributions of this work include: (i) the design and fabrication of a low-cost 8S1P $LiFePO_4$ battery system for backup power applications; (ii) the integration of an Arduino-based monitoring strategy capable of real-time voltage and current supervision; (iii) coordinated operation between the smart charger, BMS, and supervisory controller for enhanced safety and protection; (iv) the development of a modular architecture suitable for future renewable energy integration; and (v) experimental validation demonstrating stable charging behavior, reliable discharge performance, and improved operational visibility. These contributions provide a practical pathway toward intelligent, affordable, and scalable energy storage solutions for modern residential and small-scale commercial applications.

2 Materials and Method

2.1 System Architecture

The developed energy storage system consists of a 24 V, 2.5 kWh $LiFePO_4$ battery pack, an 8S BMS, a 30 A smart charger, an Arduino-based monitoring unit, and a Direct Current–Alternating Current (DC–AC) inverter. The overall system architecture is presented in Figure 1.

Figure 1 shows the interaction between the charger, battery pack, protection unit, monitoring subsystem, and inverter. The BMS performs cell balancing and protects against overcharge, over-discharge, overcurrent, short-circuit, and thermal faults. The Arduino controller acquires voltage and current measurements and displays operating information through the monitoring interface.

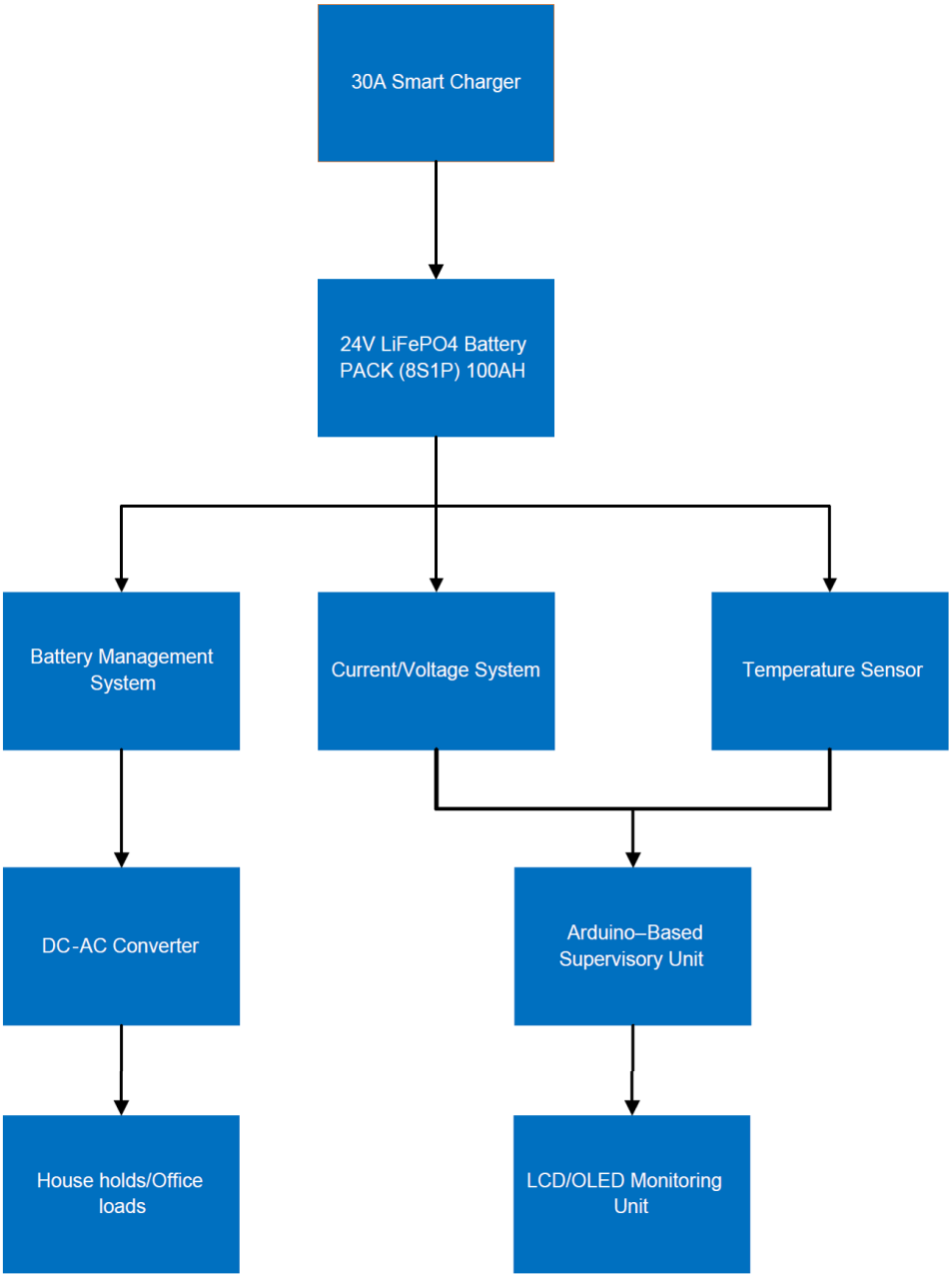


Figure 1. Block diagram of the smart lithium battery backup system

2.2 Design Procedure

The battery pack was designed using eight 3.2 V, 100 Ah $LiFePO_4$ cells connected in series to achieve a nominal voltage of 25.6 V and an energy capacity of approximately 2.56 kWh.

The charging power requirement was estimated from:

$$P = \frac{E}{t} \tag{1}$$

as described by study [10].

where P is the charging power, E is the energy rating of the battery, and t is the selected charging duration.

Battery sizing was determined using:

$$\text{Battery Capacity} = \frac{E \times \text{DOA}}{\text{DOD}} \tag{2}$$

where E is the daily energy demand, DOA is the autonomy period, and DOD is the allowable depth of discharge [11], where the depth of discharge was limited to 80% for improved cycle life.

A 600 W charging system was selected, resulting in a charging current of approximately 25 A. A full-wave rectifier and a 33,000 µF, 50 V smoothing capacitor were incorporated to improve DC output quality [12].

$$V_{peak} = V_{rms} \times \sqrt{2} \tag{3}$$

2.3 Fabrication Procedure

The fabrication process involved cell matching, busbar interconnection, terminal insulation, BMS installation, sensor integration, charger interfacing, and enclosure assembly. Figures 2–6 illustrate the major stages of construction, including battery assembly, BMS installation, charger development, system integration, and final enclosure fabrication.

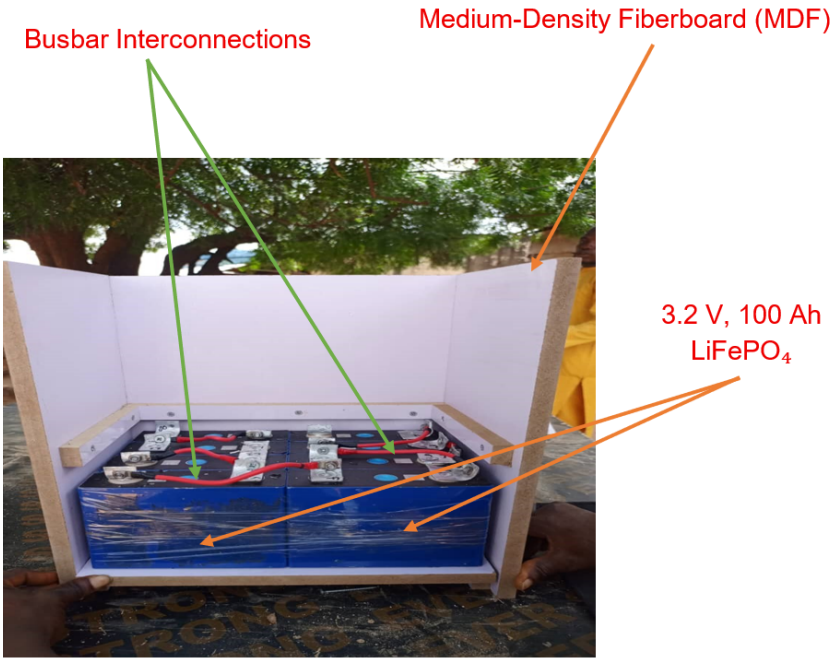
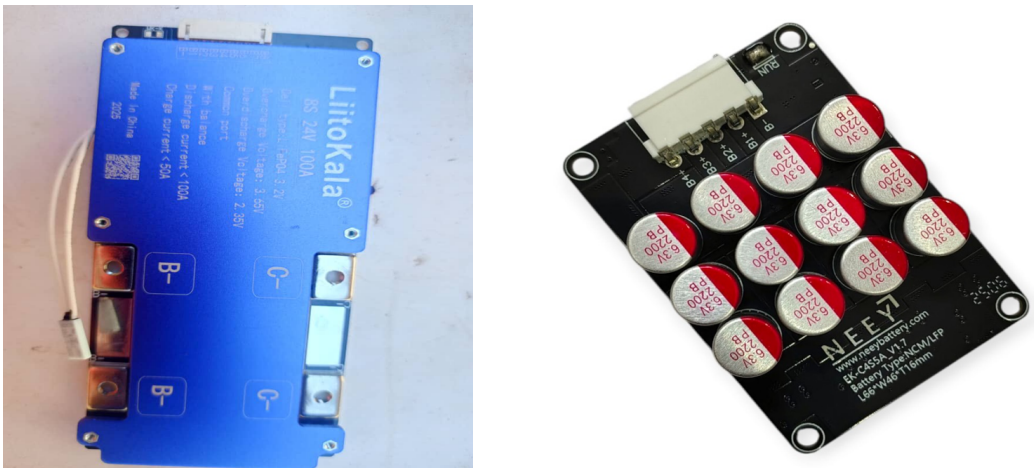


Figure 2. Lithium battery and busbar



(a) (b)

Figure 3. (a) Battery management system (BMS); (b) balancer

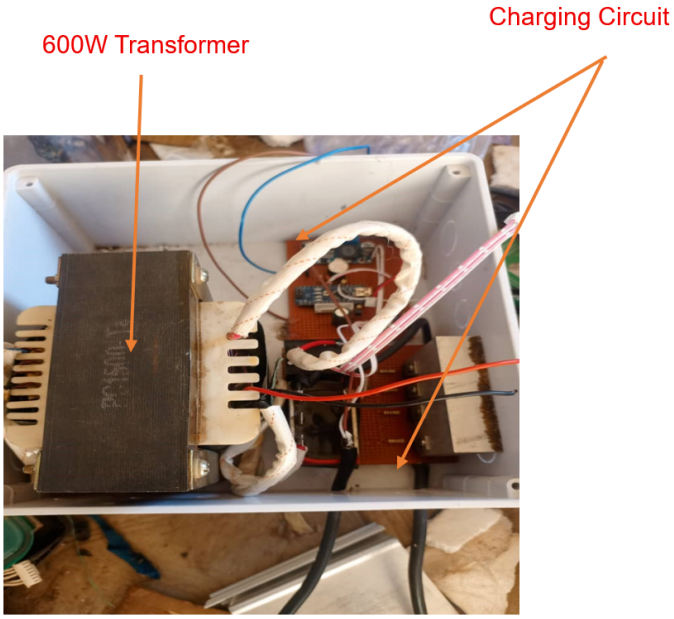


Figure 4. Battery charger

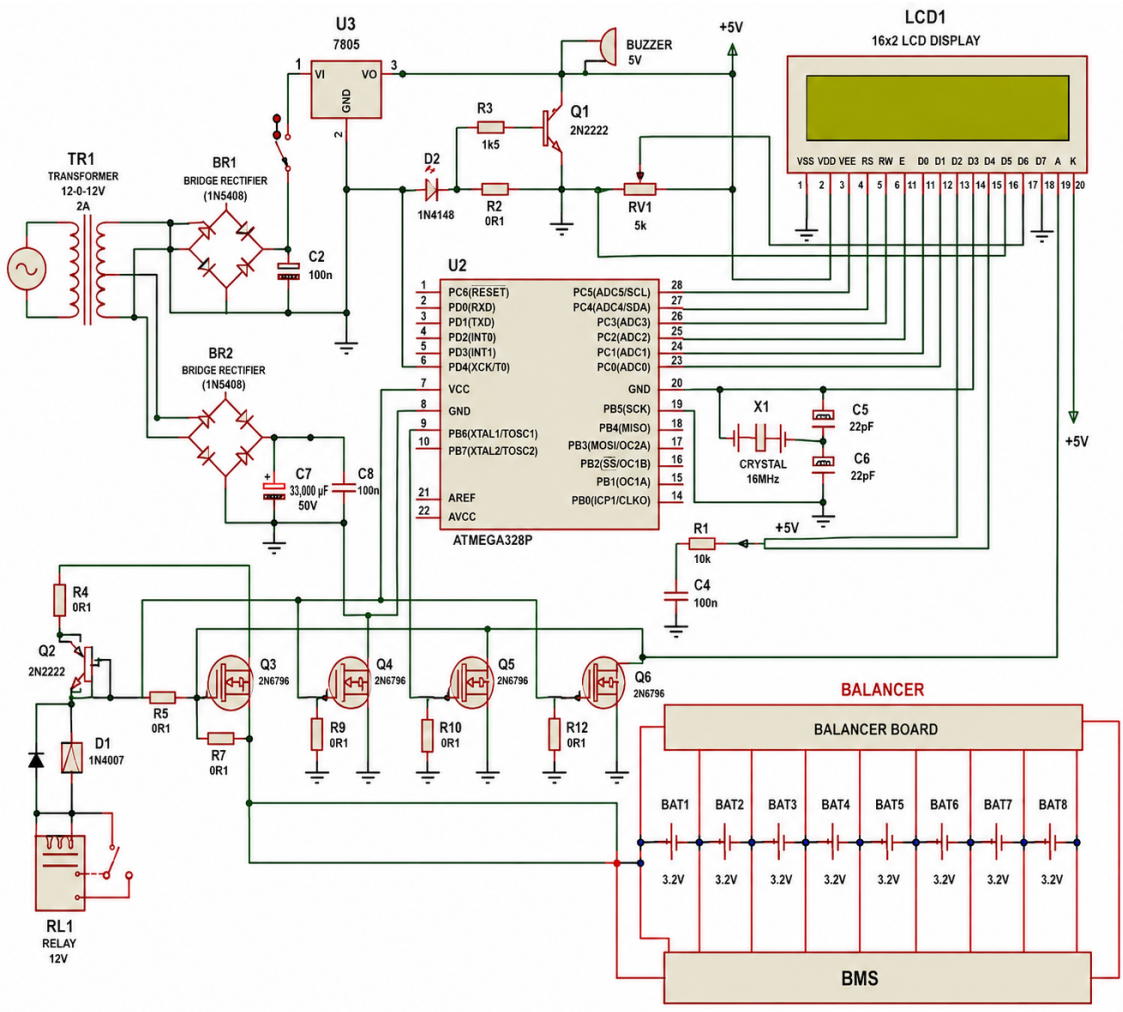


Figure 5. System circuit diagram

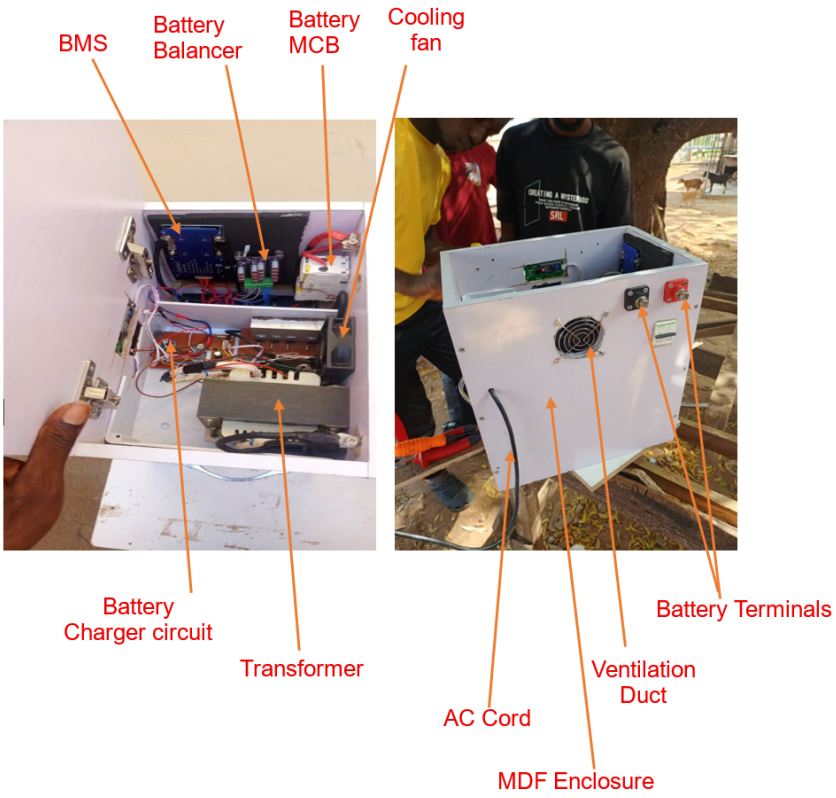


Figure 6. Design enclosure

2.4 Performance Evaluation

System performance was evaluated under controlled charging and discharging conditions. Additional metrics, including charging efficiency, discharge efficiency, round-trip efficiency, thermal behaviour, energy density, and repeatability, were assessed using Eqs. (4)–(9).

2.4.1 Charging efficiency

Charging efficiency was determined as the ratio of energy stored in the battery to the electrical energy supplied by the charger:

$$\eta_{\text{charge}} = \frac{E_{\text{stored}}}{E_{\text{input}}} \times 100\% \tag{4}$$

where:

η_{charge} = charging efficiency (%)
E_{stored} = energy stored in the battery (kWh)
E_{input} = energy supplied by the charger (kWh)

During testing, approximately 2.5 kWh of usable energy was stored from an estimated charger input energy of 2.75 kWh, resulting in a charging efficiency of approximately 91%.

2.4.2 Discharge efficiency

$$\eta_{\text{discharge}} = \frac{E_{\text{output}}}{E_{\text{stored}}} \times 100\% \tag{5}$$

The battery delivered approximately 2.4 kWh of useful output energy from a stored energy of 2.5 kWh, corresponding to a discharge efficiency of approximately 96%.

2.4.3 Round-trip efficiency

The overall round-trip efficiency represents the combined charging and discharging performance and was calculated using:

$$\eta_{\text{RT}} = \frac{E_{\text{output}}}{E_{\text{input}}} \times 100\% \tag{6}$$

The energy output was obtained to be 2.4 kWh, while the energy input was 2.75 kWh

The developed system achieved an estimated round-trip efficiency of approximately 87.3%, indicating effective energy conversion and storage performance.

2.4.4 Temperature variation during operation

Temperature measurements were recorded throughout charging and discharge cycles using the integrated monitoring system. The battery temperature remained within the safe operating range of 27 °C to 39 °C, demonstrating effective thermal stability of the $LiFePO_4$ chemistry and adequate enclosure ventilation. No thermal shutdown or overheating condition was observed during testing.

Temperature variation during operation was obtained using:

$$\Delta T = T_{\max} - T_{\min} \tag{7}$$

The battery operated within a safe thermal range of 27 °C to 39 °C, with no overheating or shutdown events, confirming stable thermal behavior.

Energy density

The gravimetric energy density of the battery pack was determined as:

$$ED = \frac{E_{\text{battery}}}{M_{\text{battery}}} \tag{8}$$

where:

ED = energy densisty of the battery sysytem (Wh/kg or kWh/kg)

E_{battery} = total energy stored in the battery (Wh or kWh)

M_{battery} = total mass (weight) of the battery system (kg)

The battery total pack weighs 22 kg, and the total energy stored in the battery was measured to be 2510 Wh when the battery is fully charged. The resulting energy density was approximately 114 Wh/kg, which is consistent with typical $LiFePO_4$ energy storage systems reported in the literature.

Repeatability and test consistency

Using the repeatability error formula:

$$\text{Repeatability Error} = \frac{X_i - \bar{X}}{\bar{X}} \times 100\% \tag{9}$$

The average energy output is:

$\bar{X}$ = 2.47 kWh

To assess operational consistency, charging and discharging tests were repeated three times under identical loading conditions. The measured energy outputs were 2.50 kWh, 2.51 kWh, and 2.40 kWh, respectively. The maximum deviation from the average output value of 2.47 kWh was approximately 2.83%, indicating good repeatability and stable system performance.

3 Results

3.1 Charging Characteristics

The battery's charging behaviour is presented in Table 1. The battery accepted a peak charging current of 23.5 A at a low state of charge, indicating strong charge acceptance capability. Battery voltage increased steadily from 24.0 V to 29.2 V during the charging period, while temperature remained within safe operating limits.

Table 1. Charging profile versus time

Time (Hours)	Voltage (V)	Current (A)	Temperature (°C)
0	24.0	23.5	27
1	25.2	23.0	29
2	26.5	22.0	31
3	27.8	18.5	34
4	28.8	10.0	37
5	29.2	2.0	39

3.2 Discharge Characteristics

Table 2 shows the discharge performance of the developed battery system. The terminal voltage decreased gradually from 29.2 V to 23.5 V over the discharge cycle, while output power remained relatively stable. The system delivered approximately 2.5 kWh of usable energy.

Table 2. Discharge voltage and power profile versus time

Time (Hours)	Voltage (V)	Power (W)
0	29.2	600
1	28.5	595
2	27.8	590
3	26.8	585
4	25.6	575
5	24.5	565
6	23.5	550

3.3 Repeatability Analysis

Three repeated discharge tests were conducted to evaluate consistency. Table 3 demonstrates close agreement among the discharge profiles. The average output energy was 2.47 kWh with a maximum deviation of approximately 2.83%.

Table 3. Comparison tests

Time (Hours)	Test 1 (V)	Test 2 (V)	Test 3 (V)
0	29.2	29.2	29.1
1	28.5	28.6	28.4
2	27.8	27.9	27.7
3	26.8	26.9	26.6
4	25.6	25.7	25.4
5	24.5	24.6	24.2
6	23.5	23.6	23.2

3.4 Performance Metrics

Experimental evaluation demonstrated stable charging and discharge behavior throughout system operation. At approximately 25% state of charge, the battery accepted up to 23.5 A during charging, confirming excellent charge absorption capability and efficient current regulation.

Table 4 summarizes the additional performance metrics obtained during experimental evaluation.

Table 4. Performance metrics

Metric	Value
Charging efficiency	91%
Discharge efficiency	96%
Round-trip efficiency	87.3%
Maximum temperature	39 °C
Minimum temperature	27 °C
Energy density	114 Wh/kg
Test repeatability error	≈2.83%

4 Discussion

The developed $LiFePO_4$ battery system demonstrated stable charging and discharge behaviour throughout experimental testing. The charging efficiency of 91% and discharge efficiency of 96% indicate effective energy transfer and low internal losses.

The measured energy density of 114 Wh/kg falls within the typical range reported for $LiFePO_4$ batteries, confirming the suitability of the selected cell chemistry for compact energy storage applications. The limited temperature variation (27–39 °C) further demonstrates the effectiveness of the BMS and enclosure ventilation strategy in maintaining thermal stability.

The repeatability error of only 2.83% confirms consistent operation and reliable performance across multiple test cycles. Such consistency is essential for backup power applications where predictable energy delivery is required.

Compared with conventional lead-acid batteries, the developed system offers higher usable depth of discharge, faster charging capability, lower maintenance requirements, and superior energy density. These advantages support its deployment in residential backup systems, solar photovoltaic installations, and small-scale microgrids.

5 Comparative Performance Analysis

Table 5 compares the developed $LiFePO_4$ battery system with conventional lead-acid technology.

Table 5. Performance comparison between conventional lead-acid and developed $LiFePO_4$ battery systems

Parameter	Lead-Acid	Developed $LiFePO_4$ System
Cycle life	200–1000 cycles	2000–6000 cycles [13]
Depth of discharge	∼50%	80–100% [13]
Energy density	30–50 Wh/kg	114 Wh/kg [14]
Charging time	8–12 h	4–5 h [15]
Maintenance	Regular	Minimal
Round-trip efficiency	70–85%	87.3%

The comparison highlights the superior operational performance and lifecycle value of the proposed system.

6 System Application Relevance

The superior performance characteristics of the $LiFePO_4$ battery system make it highly suitable for modern photovoltaic and decentralized energy systems.

In residential photovoltaic applications, the high depth of discharge and fast charging capability allow users to maximize solar energy utilization and reduce reliance on grid electricity. This improves self-consumption efficiency and enhances household energy independence.

For decentralized and off-grid systems, $LiFePO_4$ batteries provide stable voltage output, high reliability, and reduced degradation under frequent cycling. These characteristics make them suitable for rural electrification and microgrid systems where a consistent energy supply is critical.

Furthermore, the extended cycle life significantly reduces battery replacement frequency, which lowers long-term operational cost and reduces environmental waste associated with battery disposal. Literature confirms that lead-acid systems may require multiple replacements within the same period that a $LiFePO_4$ system remains operational.

In small-scale hybrid energy systems, combining photovoltaic generation with backup sources, $LiFePO_4$ batteries enhance system flexibility due to their fast charge acceptance and high round-trip efficiency. This improves energy management under variable solar irradiance conditions and supports stable hybrid operation.

Overall, the findings confirm that $LiFePO_4$-based storage systems provide superior technical performance, improved energy utilization, and stronger economic benefits compared to conventional lead-acid systems. These advantages make them a more sustainable and reliable solution for modern renewable energy integration.

7 Conclusion

A 24 V, 2.5 kWh $LiFePO_4$ battery energy storage system incorporating intelligent monitoring and protection functions was successfully designed, fabricated, and experimentally validated. The system achieved a charging efficiency of 91%, a discharge efficiency of 96%, a round-trip efficiency of 87.3%, and an energy density of 114 Wh/kg. Stable thermal performance and a repeatability error below 3% demonstrate reliable operation.

The integration of BMS protection, embedded monitoring, and smart charging provides a practical and scalable solution for residential backup power, photovoltaic energy storage, and decentralized energy applications.

Author Contributions

Conceptualization, K.M. and O.J.A.; methodology, K.M.; literature review, S.A.A., O.K.L., E.K.A., and L.O.I.; system design and integration K.M., S.J.A., and E.P.C.; validation, O.K.L., L.O.I., O.J.A., and L.O.I.; experiment, E.P.C. and O.K.L.; formal analysis, E.K.A.; investigation, K.M. and S.J.A.; resources, O.K.L. and L.O.I.; data curation, B.O.O.; writing—original draft preparation, K.M.; writing—review and editing, S.A.A., E.K.A., B.O.O., and O.J.A.; visualization, E.P.C. and S.A.A.; supervision, S.A.A. and O.J.A.; project administration, L.O.I. All

authors contributed to the development of the manuscript, reviewed the final version, and approved the manuscript for publication.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

During the preparation of this work, the authors used AI to paraphrase. Afterward, they reviewed and edited the content as necessary and took full responsibility for the publication's content.

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